170 km. However, sources say the execution of the dredging compromised bridge stability near the foundations and pillars. Poor inter-agency coordination, along with increased water flow resulting from the dredging, compounded the issue.

Incidentally, the collapse of the Aguwani-Sultanganj bridge in June last year, where a 575-metre portion of the structure crumbled into the Ganga flowing under, should have served as a wake-up call. Yet, it failed to trigger any stringent monitoring practices among engineers to prevent further incidents. Monitoring and maintenance lapses may not be the only culprit. A section of engineers tells INDIA TODAY that structural failures often stem from alterations to bridge designs and the use of substandard materials, including low-strength cement, sand and concrete, and steel rods thinner than the approved specifications. Instances of construction violations and collusive monitoring negligence have also been reported (see *Why They Crumble So Often* in accompanying graphic).

The collapse of the Aguwani-Sultanganj bridge in June last year has failed to serve as a wake-up call

Now in damage-control mode, the government has suspended 15 engineers, including 11 from the WRD and four from the rural works department, for dereliction of duty. Show-cause notices have been issued to the construction companies concerned. The chief minister has ordered the immediate implementation of a monitoring mechanism with clearly defined roles and responsibilities for all involved, aiming for both remedial action and proactive measures through a new bridge maintenance policy (see *What the State Has Done*). But mitigating the damage to the government's image? That seems to be a bridge too far right now. ■

PILLAR TO PERIL

In under a month, 12 bridges have collapsed in Bihar. It's all down to technical ineptness, bad material and collusion

Text by **AMITABH SRIVASTAVA**
Graphic by **TANMOY CHAKRABORTY**

JULY 3-4, SARAN

A portion of a 15-year-old bridge built over the Gandak river in Baniyapur block falls

Cost: Rs 7.5 lakh

Portions of a bridge made in 2004 near Janta Bazar in Lahladpur block collapse

Cost: Rs 20 lakh

A colonial-era bridge crumbles in Dandaspur Jangalvilas Tola

Current cost: Rs 10 lakh

JULY 3, SIWAN

Three small bridges/culverts collapse in the district within 24 hours, the first over a Gandak river stream in Deoria panchayat, the second over the Dhamahi river at Nautan and the third in Tewatha

Total cost: Rs 26 lakh

Reasons: Technical oversight during desilting near the foundations of old bridges and culverts under the Jal-Jeevan-Hariyali campaign's river linkage project

JUNE 30, KISHANGANJ

A pillar of a bridge over the Boond river, built with MPLAD funds in 2009-10, crumbles, causing two spans to subside by a foot

Cost: Rs 35 lakh

Reason: Non-maintenance, overloading, and illegal sand mining

JUNE 28, MADHUBANI

A 75-metre-long bridge over the Bhuthi Balan river in Madhepur, being built by the rural works department under PMGSY, comes apart

Cost: Rs 3.21 crore

Reason: A girder disintegrates

